

Last login: Thu Jul 2 10:39:50 on ttys000
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch-200mb
psql (18.4)
Type "help" for help.

tpch-200mb=# SET ROLE alice;

EXPLAIN (ANALYZE, VERBOSE, BUFFERS)

```
select
  s_name,
  s_address
from
  supplier,
  nation
where
  s_suppkey in (
    select
      ps_suppkey
    from
      partsupp
    where
      ps_partkey in (
        select
          p_partkey
        from
          part
        where
          p_name like '%ivory%'
      )
    and ps_availqty > (
      select
        0.5 * sum(l_quantity)
      from
        lineitem
      where
        l_partkey = ps_partkey
        and l_suppkey = ps_suppkey
        and l_shipdate >= date '1995-01-01'
        and l_shipdate < date '1996-01-01' + interval '1' year
      )
  )
  and s_nationkey = n_nationkey
  and n_name = 'FRANCE'
order by
  s_name;
```

-- Planning Time: 11.655 ms
-- Execution Time: 17.225 ms

SET

WARNING: Log 1: Planner Input:

DETAIL: {QUERY

```
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname supplier
        :colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
          s_acctbal" "s_comment")
      }
    :rtekind 0
    :relid 17399
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname nation
        :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
      }
    :rtekind 0
    :relid 17376
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 2
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
```

```
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 17399
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 10 11)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 17376
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9)
    :insertedCols (b)
    :updatedCols (b)
  }
)
```

```
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
    :rtindex 1
  }
  {RANGETBLREF
    :rtindex 2
  }
)
:quals
{BOOLEXP
:boolop and
:args (
  {SUBLINK
    :sublinkType 2
    :subLinkId 0
  }
  :testexpr
    {OPEXP
      :opno 96
      :opfuncid 65
    }
)
```

```

:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 129
  }
  {PARAM
    :paramkind 2
    :paramid 1
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
  }
)
:location 139
}
:operName ("=")
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
  {RANGETBLENTY
    :alias <>
    :eref
      {ALIAS
        :aliasname partsupp
        :colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_su
        pplycost" "ps_comment")
      }
    :rtekind 0
    :relid 17409
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals (
      {BOOLEXP
        :boolop and
        :args (
          {OPEXP
            :opno 521
            :opfuncid 147
            :opresulttype 16
            :opretset false
            :opcollid 0
            :inputcollid 0
            :args (
              {VAR
                :varno 1
                :varattno 3
                :vartype 23
                :vartypmod -1
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 0
                :varreturningtype 0
                :varnosyn 1
                :varattnosyn 3
                :location -1
              }
              {CONST
                :consttype 23
                :consttypmod -1
                :constcollid 0
                :constlen 4
                :constbyval true
                :constisnull false
                :location -1
                :constvalue 4 [ 100 0 0 0 0 0 0 0 ]
              }
            )
          }
        )
      }
    :location -1
  }
  {SUBLINK
    :subLinkType 0
    :subLinkId 0
    :testexpr <>
    :operName <>
    :subselect
      {QUERY
        :commandType 1
        :querySource 0
        :canSetTag true
        :utilityStmt <>
        :resultRelation 0
        :hasAggs false
        :hasWindowFuncs false
        :hasTargetSRFs false
        :hasSubLinks false
        :hasDistinctOn false
        :hasRecursive false
        :hasModifyingCTE false
        :hasForUpdate false
        :hasRowSecurity false
        :hasGroupRTE false
        :isReturn false
        :ctelist <>
        :rtable (
          {RANGETBLENTY
            :alias
              {ALIAS
                :aliasname s
                :colnames <>
              }
            :eref
              {ALIAS
                :aliasname s
                :colnames ("s_suppkey" "s_name" "s_address" "
                s_nationkey" "s_phone" "s_acctbal" "s_comment"
                )
              }
            :rtekind 0
            :relid 17399
            :inh true
            :relkind r
            :rellockmode 1

```

s_suppkey IN (SELECT ps_suppkey FROM ...)

ps_availqty > 100

```

:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17399
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
)
{VAR
:varno 1
:varattno 2
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
}
:location -1
}
)
{OEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:varatype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
)
{FUNCEXP
:funcid 1740
:funcresulttype 1700
:funcset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
:location -1
}
)
:location -1
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 ]
}
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>

```

s.s_suppkey =
partsupp.ps_suppkey

s.s_acctbal > 0

```

:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
)
:location -1
}
)
}
)
:rtperminfos {
{RTEPERMISSIONINFO
:relid 17409
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
}
:jointree
{FROMEXPR
{fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals
{BOOLEXPR
:boolop and
:args (
{SUBLINK
:subLinkType 2
:subLinkId 0
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 302
}
{PARAM
:paramkind 2
:paramid 1
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
}
:location 313
}
}
:operName ("=")
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTRY
:alias <>
:eref
{ALIAS
:aliasname part
:colnames ("p_partkey" "p_name" "p_mfg" "p_b
rand" "p_type" "p_size" "p_container" "p_reta
ilprice" "p_comment")
}
}
:rtkind 0
:relid 17387
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rtperminfos {
{RTEPERMISSIONINFO
:relid 17387
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
}
:jointree
{FROMEXPR
{fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals
{OPEXPR
:opno 1209
:opfuncid 850
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100

```

ps_partkey in (

```

:args (
  {RELABELTYPE
    :arg
    {VAR
      :varno 1
      :varattno 2
      :vartype 1043
      :vartypmod 59
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 2
      :location 567
    }
    :resulttype 25
    :resulttypmod -1
    :resultcollid 100
    :relabelformat 2
    :location -1
  }
  {CONST
    :consttype 25
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location 579
    :constvalue 11 [ 44 0 0 0 37 105 118 111 1
14 121 37 ]
  }
)
:location 574
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location 397
    }
    :resno 1
    :resname p_partkey
    :ressortgroupref 0
    :resorigtbl 17387
    :resorigcol 1
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returninglist <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
)
:location 313
}
{OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {FUNCEXPR
    :funcid 1740
    :funcresulttype 1700
    :funcresultset false
    :funcvariadic false
    :funcformat 2
    :funccollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 3
        :location 643
      }
    )
  }
)
:location -1
}
{SUBLINK
:sublinkType 4
:subLinkId 0
:testexpr <>
:operName <>
:subselect
  {QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
    :utilityStmt <>
    :resultRelation 0
    :hasAggs true
    :hasWindowFuncs false
    :hasTargetSRFs false
    :hasSublinks false
    :hasDistinctOn false
    :hasRecursive false
    :hasModifyingCTE false
    :hasForUpdate false
    :hasRowSecurity false
    :hasGroupRTe false
    :isReturn false
    :ctelist <>
    :rtable (
      {RANGETBENTRY
        :alias <>

```

p_name like '%ivory%'

ps_availqty > (

l_partkey = ps_partkey

and l_suppkey =
ps_suppkey

and l_shipdate >= date
'1995-01-01'

and l_shipdate < date
'1995-01-01' + interval
'1' year

```

:eref
{
  ALIAS
  :aliasname lineitem
  :colnames ("l_orderkey" "l_partkey" "l_sup
pkey" "l_linenumber" "l_quantity" "l_exten
dedprice" "l_discount" "l_tax" "l_returnfl
ag" "l_linestatus" "l_shipdate" "l_commitd
ate" "l_receiptdate" "l_shipinstruct" "l_s
hipmode" "l_comment")
}
}
:rtekind 0
:reldid 17440
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
(RTEPERMISSIONINFO
:reldid 17440
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 9 10 12 18)
:insertedCols (b)
:updatedCols (b)
)
)
:jointree
{FROMEXPR
:fromlist {
(RANGEtblref
:rtindex 1
)
}
}
:quals
{BOOLEXPR
:boolop and
:args {
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args {
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 924
}
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 936
}
}
}
:location 934
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args {
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 991
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 1003
}
}
}
:location 1001
}
{OPEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args {
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 1058
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 1077
:constvalue 4 [ -34 -8 -1 -1 -1 -

```

```

        1 -1 ]
    }
}
:location 1069
}
{OEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 1134
}
{OEXPR
:opno 1076
:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 1152
:constvalue 4 [ -34 -8 -1 -1 -1 -
1 -1 -1 ]
}
{CONST
:consttype 1186
:consttypmod 327679
:constcollid 0
:constlen 16
:constbyval false
:constisnull false
:location 1176
:constvalue 16 [ 0 0 0 0 0 0 0 0
0 0 0 0 12 0 0 0 ]
}
}
:location 1165
}
}
:location 1145
}
}
:location 987
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{OEXPR
:opno 1760
:opfuncid 1726
:opresulttype 1700
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location 738
:constvalue 8 [ 32 0 0 0 -1 -128 -120 1
9 ]
}
{AGGREF
:aggfnoid 2114
:aggtype 1700
:aggcollid 0
:inputcollid 0
:aggtransno 0
:aggargtypes (o 1700)
:aggdirectargs <>
:args (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 748
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar false
:aggvariadic false
:aggkind n
:aggresorted false
:agglevelsup 0
:aggssplit 0
:aggno -1
:aggtransno -1
:location 744
}
}
:location 742
}
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
)
:override 0

```

```

        :onConflict <>
        :returningOldAlias <>
        :returningNewAlias <>
        :returningList <>
        :groupClause <>
        :groupDistinct false
        :groupingSets <>
        :havingQual <>
        :windowClause <>
        :distinctClause <>
        :sortClause <>
        :limitOffset <>
        :limitCount <>
        :limitOption 0
        :rowMarks <>
        :setOperations <>
        :constraintDeps <>
        :withCheckOptions <>
        :stmt_location 0
        :stmt_len 0
    }
    :location 657
}
}
:location 655
}
)
:location 639
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
    :expr
    (VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location 191
    )
    :resno 1
    :resname ps_supkey
    :ressortgroupref 0
    :resorigtbl 17409
    :resorigcol 2
    :resjunk false
    }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 139
}
{OPEXPR
:opno 96
:opfncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
    :varno 1
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 4
    :location 1233
    }
    {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location 1247
    }
    )
:location 1245
}
{OPEXPR
:opno 1054
:opfncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
    {VAR
    :varno 2
    :varattno 2
    :vartype 1042
    :vartypmod 29
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 2
    :location 1271
    }
    {CONST
    :consttype 1042
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location 1280
    :constvalue 10 [ 40 0 0 0 70 82 65 78 67 69 ]

```

s_nationkey =
n_nationkey
and n_name =
'FRANCE'


```

    }
    }
    :location 1278
  }
}
:location 1229
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 51
}
:resno 1
:resname s_name
:resortgroupref 1
:resorigtbl 17399
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 1043
:vartypmod 44
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 67
}
:resno 2
:resname s_address
:resortgroupref 0
:resorigtbl 17399
:resorigcol 3
:resjunk false
}
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: max_parallel_hazard_walker: Param paramid=1 in_safe_list=0 → reason=outer_reference
WARNING: max_parallel_hazard_walker: Param paramid=1 in_safe_list=0 → reason=outer_reference
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 propparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 propparallel=s (s=safe r=restricted u=unsafe)
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtkind 0
:relid 17440
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromC1 true
:securityQuals <>
}
)
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17440
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 9 10 12 18)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointree
{FROMEXPR

```

```

:fromlist (
  {RANGETBLREF
   :rtindex 1
  }
)
:quals (
  {OEXPR
   :opno 96
   :opfuncid 65
   :opresulttype 16
   :opretset false
   :opcollid 0
   :inputcollid 0
   :args (
     {VAR
      :varno 1
      :varattno 2
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 2
      :location 924
     }
     {PARAM
      :paramkind 1
      :paramid 0
      :paramtype 23
      :paramtypmod -1
      :paramcollid 0
      :location 936
     }
   )
   :location 934
 }
  {OEXPR
   :opno 96
   :opfuncid 65
   :opresulttype 16
   :opretset false
   :opcollid 0
   :inputcollid 0
   :args (
     {VAR
      :varno 1
      :varattno 3
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 3
      :location 991
     }
     {PARAM
      :paramkind 1
      :paramid 1
      :paramtype 23
      :paramtypmod -1
      :paramcollid 0
      :location 1003
     }
   )
   :location 1001
 }
  {OEXPR
   :opno 1098
   :opfuncid 1090
   :opresulttype 16
   :opretset false
   :opcollid 0
   :inputcollid 0
   :args (
     {VAR
      :varno 1
      :varattno 11
      :vartype 1082
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 11
      :location 1058
     }
     {CONST
      :consttype 1082
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location 1077
      :constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 -1 ]
     }
   )
   :location 1069
 }
  {OEXPR
   :opno 2345
   :opfuncid 2338
   :opresulttype 16
   :opretset false
   :opcollid 0
   :inputcollid 0
   :args (
     {VAR
      :varno 1
      :varattno 11
      :vartype 1082
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 11
      :location 1134
     }
     {CONST
      :consttype 1114
      :consttypmod -1
      :constcollid 0
      :constlen 8
      :constbyval true
      :constisnull false
      :location -1
      :constvalue 8 [ 0 32 -39 -79 49 -115 -1 -1 ]
     }
   )
   :location 1145
 }
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
   :expr
   {OEXPR
    :opno 1760

```

```

:opfuncid 1726
:opresulttype 1700
:opreset false
:opcollid 0
:inputcollid 0
:args (
  (CONST
    :consttype 1700
    :consttypmod -1
    :constcollid 0
    :constlen -1
    :constbval false
    :constisnull false
    :location 738
    :constvalue 8 [ 32 0 0 0 -1 -128 -120 19 ]
  )
  (AGGREF
    :aggfnoid 2114
    :aggtype 1700
    :aggcollid 0
    :inputcollid 0
    :aggtranstype 0
    :aggargtypes (o 1700)
    :aggdirectargs <>
    :args (
      (TARGETENTRY
        :expr
        {VAR
          :varno 1
          :varattno 5
          :vartype 1700
          :vartypmod 983046
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 5
          :location 748
        }
      )
      :resno 1
      :resname <>
      :ressortgroupref 0
      :resorigtbl 0
      :resorigcol 0
      :resjunk false
    )
  )
  :aggorder <>
  :aggdistinct <>
  :aggfilter <>
  :aggstar false
  :aggvariadic false
  :aggkind n
  :aggpresorted false
  :agglevelsup 0
  :aggssplit 0
  :aggno -1
  :aggtransno -1
  :location 744
)
:location 742
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
)
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
)

```

```

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 1098 :opfuncid 1098 :opresulttype 16 :opreset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 11 :vartype 1
082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 11 :location -1) (CONST :consttype 1082 :consttypmod -1 :constcollid 0 :constlen 4 :constb
val true :constisnull false :location -1 :constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 -1 ]) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof
false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_
serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_buck
etsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICTINFO :clause {OPEXPR :opno 2345 :opfuncid 2338 :opresulttype 16 :opreset false :opcollid 0 :inputco
llid 0 :args ((VAR :varno 1 :varattno 11 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 11 :location -1) (CONST :consttype 1114
:consttypmod -1 :constcollid 0 :constisnull false :location -1 :constvalue 8 [ 0 32 -39 -79 49 -115 -1 -1 ]) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false
:is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b
1) :right_rels (b) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opr
esulttype 16 :opreset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varat
tnosyn 2 :location -1) (PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false
:is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause
<> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1)}) :right_em (EQUIVALENCEMEMBER :em_expr (PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opreset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) (PARAM :paramkind 1 :paramid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1)}) :right_em (EQUIVALENCEMEMBER :em_expr (PARAM :paramkind 1 :paramid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0)
WARNING: max_parallel_hazard_walker: Param paramid=0 in safe_list=0 + reason=outer reference

```

```

WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 1098 :opfuncid 1098 :opresulttype 16 :opreset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 11 :vartype
1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 11 :location -1) (CONST :consttype 1082 :consttypmod -1 :constcollid 0 :constlen 4 :constb
val true :constisnull false :location -1 :constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 -1 ]) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproo
f false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b 1) :left_rels (b) :orclause <> :rinfo_
serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_buc
ketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICTINFO :clause {OPEXPR :opno 2345 :opfuncid 2338 :opresulttype 16 :opreset false :opcollid 0 :inputco
llid 0 :args ((VAR :varno 1 :varattno 11 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 11 :location -1) (CONST :consttype 1114
:consttypmod -1 :constcollid 0 :constisnull false :location -1 :constvalue 8 [ 0 32 -39 -79 49 -115 -1 -1 ]) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false
:is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause
<> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opr
esulttype 16 :opreset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varat
tnosyn 2 :location -1) (PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false
:is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause
<> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1)}) :right_em (EQUIVALENCEMEMBER :em_expr (PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opreset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) (PARAM :paramkind 1 :paramid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1)}) :right_em (EQUIVALENCEMEMBER :em_expr (PARAM :paramkind 1 :paramid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0) | parallel_safe = false | reason = PROPARALLEL_REST

```

```
ICTED
WARNING: is_parallel_safe: INPUT qual = ({OEXPR :opno 1760 :opfuncid 1726 :opresulttype 1700 :opretset false :opcollid 0 :inputcollid 0 :args ({CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 8 [ 32 0 0 0 -1 -128 -120 19 ]}) (AGGREF :aggfnoid 2114 :aggtype 1700 :aggcollid 0 :inputcollid 0 :aggtranstype 2281 :aggargtypes (o 1700) :aggdirectargs <> :args ({TARGETENTRY :expr {VAR :varno 1 :varattno 5 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1} :resno 1 :resname <> :ressortgroupref 0 :resorigtbl 0 :resorigcol 0 :resjunk false}) :aggorder <> :aggdistinct <> :aggfilter <> :aggstar false :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtransno 0 :location -1}) :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({OEXPR :opno 1760 :opfuncid 1726 :opresulttype 1700 :opretset false :opcollid 0 :inputcollid 0 :args ({CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 8 [ 32 0 0 0 -1 -128 -120 19 ]}) (AGGREF :aggfnoid 2114 :aggtype 1700 :aggcollid 0 :inputcollid 0 :aggtranstype 2281 :aggargtypes (o 1700) :aggdirectargs <> :args ({TARGETENTRY :expr {VAR :varno 1 :varattno 5 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn n 5 :location -1} :resno 1 :resname <> :ressortgroupref 0 :resorigtbl 0 :resorigcol 0 :resjunk false}) :aggorder <> :aggdistinct <> :aggfilter <> :aggstar false :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtransno 0 :location -1}) :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 5 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 5 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
}
:rtekind 0
:reliid 17399
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:reliid 17399
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 0 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 2
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
}
:location -1
)
{OEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 6 [ 24 0 0 0 0 -128 ]
}
}
:location -1
}
)
}
:mergeActionList <>
:mergeTargetRelation 0
```

```

:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
:windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:smt_location -1
:smt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 17
00 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :cons
tbval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof fals
e :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_seria
l 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsiz
e -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :a
rgs ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 2 :para
mtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :securit
y_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1
:eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b)
:varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}) :right_em (E
QUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 2 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOIND
OMAIN :jd_relds (b 1)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 1
700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :con
stbval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof fals
e :has_volatile :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :right_relds (b) :orclause <> :rinfo_seri
al 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsiz
e -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :
args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 2 :par
amtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :securi
ty_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1
:eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b)
:varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}) :right_em (E
QUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 2 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOIN
DOMAIN :jd_relds (b 1)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel
_safe = false | reason = PROPARALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilitySmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:table (
{RANGETBLENTY
:alias
{ALIAS
:aliasname s
:colnames <>
}
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtelkind 0
:reld 17399
:inh false
:relkind r
:rellockmode 1
:permindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rtelperminfos (
{RTEPERMISSIONINFO
:reld 17399
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 0 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbval false
:constisnull false
:location -1
:constvalue 6 [ 24 0 0 0 0 -128 ]
}
}
}
}

```

```

    }
  }
  :location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING:  is_parallel_safe: INPUT qual = ({RESTRICINFO :clause {OPEXPR :opno 1756 :opfucind 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 17
00 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :cons
tbyval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof fals
e :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_seria
l 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsiz
e -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING:  is_parallel_safe: OUTPUT qual = ({RESTRICINFO :clause {OPEXPR :opno 1756 :opfucind 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 1
700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :con
stbyval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof fal
se :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_seri
al 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsi
ze -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)
WARNING:  is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)
WARNING:  is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = <>
WARNING:  is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  Log 2 : After pull up At starting of Path Generation:
DETAIL:  {QUERY
  :commandType 1
  :querySource 0
  :canSetTag true
  :utilityStmt <>
  :resultRelation 0
  :hasAggs false
  :hasWindowFuncs false
  :hasTargetSRFs false
  :hasSubLinks true
  :hasDistinctOn false
  :hasRecursive false
  :hasModifyingCTE false
  :hasForUpdate false
  :hasRowSecurity true
  :hasGroupRTe false
  :isReturn false
  :codelist <>
  :table (
    {RANGETBENTRY
      :alias <>
      :eref
      {ALIAS
        :aliasname supplier
        :colnames ("s_suppkkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
      }
      :rtekind 0
      :relid 17399
      :inh false
      :relkind r
      :rellockmode 1
      :perminfoindex 1
      :tablesample <>
      :lateral false
      :inFromCl true
      :securityQuals <>
    }
    {RANGETBENTRY
      :alias <>
      :eref
      {ALIAS
        :aliasname nation
        :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
      }
      :rtekind 0
      :relid 17376
      :inh false
      :relkind r
      :rellockmode 1
      :perminfoindex 2
      :tablesample <>
      :lateral false
      :inFromCl true
      :securityQuals <>
    }
    {RANGETBENTRY
      :alias
      {ALIAS
        :aliasname ANY_subquery
        :colnames <>
      }
      :eref
      {ALIAS
        :aliasname ANY_subquery
        :colnames ("ps_suppkkey")
      }
    }
  )
}

```

```

:rtekind 1
:subquery <>
:security_barrier false
:relid 0
:inh false
:relkind <>
:rellockmode 0
:perminfoindex 0
:lateral false
:inFromCl false
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname partsupp
:colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "p
s_comment")
}
:rtekind 0
:relid 17409
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals ((
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 3
:location -1
}
){CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [(100 0 0 0 0 0 0 0)]
}
)
:location -1
}
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:sublinktype 0
:testexpr <>
:paramids <>
:plan_id 2
:plan_name SubPlan\ 2
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 2)
:args (
{VAR
:varno 4
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location -1
}
)
:startup_cost 0
:per_call_cost 110
}
{SUBPLAN
:sublinktype 2
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location -1
}
){PARAM
:paramkind 1
:paramid 3
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
:paramids (i 3)
:plan_id 3
:plan_name SubPlan\ 3
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 117.5025
:per_call_cost 0.0025
}

```

```

    )
  }
})
}
{RANGETBLENTRY
:alias
{ALIAS
:aliasname ANY_subquery
:colnames <>
}
:eref
{ALIAS
:aliasname ANY_subquery
:colnames ("p_partkey")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 0
:inh false
:relkind <>
:rellockmode 0
:perminfoindex 0
:lateral false
:inFromCl false
:securityQuals <>
}
{RANGETBLENTRY
:alias <>
:eref
{ALIAS
:aliasname part
:colnames ("p_partkey" "p_name" "p_mfg" "p_brand" "p_type" "p_size"
"p_container" "p_retailprice" "p_comment")
}
:rtekind 0
:relid 17387
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17399
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17376
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17409
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17387
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
}
)
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
)
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 1233
}
)
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 1247
}
)
:location 1245
}
)
{OEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR

```

supplier.s_nationkey =
nation.n_nationkey


```

:varno 2
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location 1271
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 1280
:constvalue 10 [ 40 0 0 0 70 82 65 78 67 69 ]
}
}
:location 1278
}
}
}
:rarg
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 4
}
)
:quals (
{OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{FUNCEXPR
:funcid 1740
:funcresulttype 1700
:funcset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 3
:location 643
}
)
:location -1
}
{SUBPLAN
:sublinkType 4
:testexpr <>
:paramids <>
:plan_id 1
:plan_name SubPlan\ 1
:firstColType 1700
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEnFalse false
:parallel_safe false
:setParam <>
:parParam (i 0 1)
:args (
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 936
}
)
{VAR
:varno 4
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location 1003
}
)
:startup_cost 0
:per_call_cost 8.469999999999999
}
}
:location 655
}
}
}
:rarg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 6
}
)
:quals (
{OPEXPR
:opno 1209
:opfuncid 850
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{RELABELTYPE
:arg
{VAR
:varno 6
:varattno 2
:vartype 1043
:vartypmod 59
:varcollid 100
:varnullingrels (b)
:varlevelsup 0

```

n_name = 'FRANCE'

```

        :varreturningtype 0
        :varnosyn 6
        :varattnosyn 2
        :location 567
    }
    :resulttype 25
    :resulttypmod -1
    :resultcollid 100
    :relabelformat 2
    :location -1
    }
    {CONST
    :consttype 25
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location 579
    :constvalue 11 [ 44 0 0 0 37 105 118 111 114 121 37 ]
    }
    }
    :location 574
    }
    )
    }
    :usingClause <>
    :join_using_alias <>
    :quals (
    {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 4
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 1
    :location 302
    }
    {VAR
    :varno 6
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 6
    :varattnosyn 1
    :location 397
    }
    )
    :location 313
    }
    )
    :alias <>
    :rtindex 0
    }
    :usingClause <>
    :join_using_alias <>
    :quals (
    {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 129
    }
    {VAR
    :varno 4
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 2
    :location 191
    }
    )
    :location 139
    }
    )
    :alias <>
    :rtindex 0
    }
    )
    :quals <>
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 2
    :vartype 1042
    :vartypmod 29
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location 51
    }
    }
    :resno 1
    :resname s_name
    :ressortgroupref 1
    :resorigtbl 17399
    :resorigcol 2
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 3

```

p_name LIKE
'%ivory%'

ps_partkey = p_partkey

s_suppkey =
ps_suppkey

rtypmod -1 :varcolldid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcolldid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 2 :security_level 1 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_serial 9 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selec 0.04 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcolldid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 4 6))} :right_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcolldid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 4 6))} :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96)}
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause (OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcolldid 0 :inputcolldid 0 :args ((VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcolldid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcolldid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 2 :security_level 1 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_serial 9 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selec 0.04 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcolldid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 4 6))} :right_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcolldid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 4 6))} :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96)) | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 2 :vartype 1042 :vartypmod 29 :varcolldid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1} {VAR :varno 1 :varattno 3 :vartype 1043 :vartypmod 44 :varcolldid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcolldid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 2 :security_level 1 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_serial 9 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selec 0.04 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcolldid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 4 6))} :right_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcolldid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 4 6))} :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96)) | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 2 :vartype 1042 :vartypmod 29 :varcolldid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1} {VAR :varno 1 :varattno 3 :vartype 1043 :vartypmod 44 :varcolldid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcolldid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 2 :security_level 1 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_serial 9 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selec 0.04 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcolldid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 4 6))} :right_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcolldid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 4 6))} :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96)) | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 2 :vartype 1042 :vartypmod 29 :varcolldid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1} {VAR :varno 1 :varattno 3 :vartype 1043 :vartypmod 44 :varcolldid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcolldid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 2 :security_level 1 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_serial 9 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selec 0.04 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcolldid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 4 6))} :right_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcolldid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 4 6))} :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96)) | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 2 :vartype 1042 :vartypmod 29 :varcolldid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1} {VAR :varno 1 :varattno 3 :vartype 1043 :vartypmod 44 :varcolldid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcolldid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 2 :security_level 1 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_serial 9 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selec 0.04 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcolldid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 4 6))} :right_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcolldid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2 4 6))} :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96)) | parallel_safe = true | reason = PROPAGALLEL_SAFE
n = PROPAGALLEL_SAFE

QUERY PLAN

Sort (cost=697379.06..697379.26 rows=80 width=52) (actual time=412.367..412.376 rows=65.00 loops=1)
Output: **supplier.s_name, supplier.s_address**
Sort Key: supplier.s_name
Sort Method: quicksort Memory: 29kB
Buffers: shared hit=6526 read=4445
-> Nested Loop **Semi Join** (cost=1821.17..697376.53 rows=80 width=52) (actual time=26.494..412.016 rows=65.00 loops=1)
Output: supplier.s_name, supplier.s_address
Buffers: shared hit=6523 read=4445
-> Nested Loop (cost=4.90..97.57 rows=80 width=56) (actual time=0.233..0.356 rows=78.00 loops=1)
Output: supplier.s_name, supplier.s_address, supplier.s_supkey
Buffers: shared hit=38 read=2
-> Seq Scan on public.nation (cost=0.00..1.31 rows=1 width=4) (actual time=0.024..0.032 rows=1) (loops=1)
Output: nation.n_nationkey, nation.n_name, nation.n_regionkey, nation.n_comment
Filter: (**nation.n_name = 'FRANCE'**::bpchar)
Rows Removed by Filter: 24
Buffers: shared hit=1
-> Bitmap Heap Scan on public.supplier (cost=4.90..95.45 rows=80 width=60) (actual time=0.206..0.301 rows=78.00 loops=1)
Output: supplier.s_supkey, supplier.s_name, supplier.s_address, supplier.s_nationkey, supplier.s_phone, supplier.s_acctbal, supplier.s_comment
Recheck Cond: (**supplier.s_nationkey = nation.n_nationkey**)
Heap Blocks: exact=37
Buffers: shared hit=37 read=2
-> Bitmap Index Scan on idx_supplier_nationkey (cost=0.00..4.88 rows=80 width=0) (actual time=0.175..0.178 rows=78.00 loops=1)
Index Cond: (supplier.s_nationkey = nation.n_nationkey)
Index Searches: 1
Buffers: shared read=2
-> **Hash Semi Join** (cost=5.27..11143.28 rows=1 width=4) (actual time=5.277..5.277 rows=0.83 loops=78)
Output: partsupp.ps_supkey
Hash Cond: (**partsupp.ps_partkey = part.p_partkey**)
Buffers: shared hit=6485 read=4443
-> **Index Scan using idx_partsupp_supkey on public.partsupp** (cost=0.29..10127.25 rows=13 width=8) (actual time=0.819..5.094 rows=15.64 loops=78)
Output: partsupp.ps_partkey, partsupp.ps_supkey, partsupp.ps_availqty, partsupp.ps_supplycost, partsupp.ps_comment
Index Cond: (partsupp.ps_supkey = supplier.s_supkey)
Filter: (**(partsupp.ps_availqty > 100)**) AND (ANY (partsupp.ps_supkey = (hashed SubPlan 3).col1)) AND ((partsupp.ps_availqty)::numeric > (SubPlan 1)))
Rows Removed by Filter: 17
Index Searches: 78
Buffers: shared hit=6291 read=3875
SubPlan 3
-> Seq Scan on public.supplier s (cost=0.00..113.00 rows=1801 width=4) (actual time=0.017..1.230 rows=1802.00 loops=1)
Output: s.s_supkey
Filter: (**(s.s_acctbal > 0)::numeric**)
Rows Removed by Filter: 198
Buffers: shared hit=88
SubPlan 1
-> Aggregate (cost=8.46..8.47 rows=1 width=32) (actual time=0.140..0.140 rows=1.00 loops=1791)
Output: (**0.5 * sum(lineitem.l_quantity)**)
Buffers: shared hit=5110 read=2280
-> Index Scan using idx_lineitem_partkey_supkey_shipdate on public.lineitem (cost=0.43..8.45 rows=1 width=5) (actual time=0.111..0.139 rows=1.12 loops=1791)
Output: lineitem.l_orderkey, lineitem.l_partkey, lineitem.l_supkey, lineitem.l_linenum, lineitem.l_quantity, lineitem.l_extendedprice, lineitem.l_discount, lineitem.l_tax, lineitem.l_returnflag, lineitem.l_linestatus, lineitem.l_shipdate, lineitem.l_commitdate, lineitem.l_receiptdate, lineitem.l_shipinstruct, lineitem.l_shipmode, lineitem.l_comment
Index Cond: ((**lineitem.l_partkey = partsupp.ps_partkey**) AND (**lineitem.l_supkey = partsupp.ps_supkey**) AND (**lineitem.l_shipdate >= '1995-01-01'::date**) AND (**lineitem.l_shipdate < '1996-01-01 00:00:00'::timestamp without time zone**))
Index Searches: 1791
Buffers: shared hit=5110 read=2280
-> Hash (cost=965.48..965.48 rows=4040 width=4) (actual time=13.677..13.677 rows=2082.00 loops=1)
Output: **part.p_partkey**
Buckets: 4096 Batches: 1 Memory Usage: 106kB
Buffers: shared hit=194 read=568
-> Bitmap Heap Scan on public.part (cost=50.98..965.48 rows=4040 width=4) (actual time=2.268..13.284 rows=2082.00 loops=1)
Output: part.p_partkey
Recheck Cond: (**(part.p_name)::text ~ '%Ivory%':::text**)
Heap Blocks: exact=752
Buffers: shared hit=194 read=568
-> Bitmap Index Scan on idx_part_name_trgm (cost=0.00..49.97 rows=4040 width=0) (actual time=1.639..1.639 rows=2082.00 loops=1)
Index Cond: (**(part.p_name)::text ~ '%Ivory%':::text**)
Index Searches: 1
Buffers: shared hit=7 read=3
Planning:
Buffers: shared hit=358 read=44
Planning Time: 24.567 ms
Execution Time: 413.689 ms
(68 rows)
tpch-200mb=>